prominent Academy

tcs

DATA ENGINEER

PySpark

scenario-based interview questions

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Question:

Your company processes real-time transactions, but duplicates appear due to retries.

1️⃣ Use Watermarking and Deduplication

```python
from pyspark.sql.functions import col

streaming_df =
spark.readStream.format("delta").load("s3://transactions")

deduplicated_stream = (
    streaming_df
    .withWatermark("event_time", "10 minutes")
    .dropDuplicates(["transaction_id"])
)

deduplicated_stream.writeStream.format("delta").start("s3://cl
eaned_transactions")
```

✅ Removes duplicate records in real-time.

Common Mistakes:

❌ Not setting a proper watermark, leading to data loss or retention issues.

❌ Using batch deduplication instead of streaming deduplication.

Question:

You need to detect repeated fraud attempts within a short time window.
Solution:

1️⃣ Use Sliding Windows to Identify Rapid Transactions

```python
from pyspark.sql.functions import window, count

fraud_attempts = (
    streaming_df
    .groupBy(window("event_time", "10 minutes", "5 minutes"),
"user_id")
    .agg(count("*").alias("txn_count"))
    .filter(col("txn_count") > 5)  # More than 5 transactions
in 10 mins
)

fraud_attempts.writeStream.format("console").start()
```

✅ Detects rapid transaction attempts, a sign of fraud.

Common Mistakes:
❌ Using batch aggregation instead of streaming aggregation.
❌ Setting a static window size instead of a sliding window.

Question:

Your PySpark joins are slow, causing high shuffle costs.

Solution:
1️⃣ Use Broadcast Joins for Small Datasets

```python
from pyspark.sql.functions import broadcast

df_large.join(broadcast(df_small), "id")
```

✅ Eliminates costly shuffling operations.

Common Mistakes:

❌ Broadcasting large tables, causing memory overflow.
❌ Not monitoring query plans to check join strategies.

Question:

Your joins on large tables are slow due to high shuffle costs.

Solution:
1️⃣ Use Bloom Filters to Optimize Joins

```python
from pyspark.sql.functions import col

spark.conf.set("spark.sql.optimizer.dynamicPartitionPruning.en
abled", "true")

df_large = spark.read.parquet("s3://data/large_table")
df_small = spark.read.parquet("s3://data/small_table")

df_small = df_small.select("user_id").dropDuplicates()
df_large =
df_large.filter(col("user_id").isin(df_small.collect()))
```

✅ Speeds up joins by reducing unnecessary data scans.

Common Mistakes:
❌ Not enabling dynamic partition pruning, causing full table scans.
❌ Using .collect() on large datasets, leading to memory issues.

Question:

Your fraud detection model needs to identify unusual spending behavior over time.

Solution:
1️⃣ Use Moving Averages for Fraud Detection

python

```
from pyspark.sql.window import Window
from pyspark.sql.functions import avg

window_spec =
Window.partitionBy("user_id").orderBy("transaction_time").rows
Between(-5, 0)

df = df.withColumn("avg_spend",
avg("amount").over(window_spec))
df = df.withColumn("is_anomalous", (df["amount"] > 2 *
df["avg_spend"]).cast("int"))
```

✅ Flags transactions that are significantly above normal spending.

Common Mistakes:
❌ Not using a moving window, leading to inaccurate anomaly detection.
❌ Applying global averages instead of per-user behavior analysis.

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